SHUBHAM KUMAR

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Education

Indian Institute of Technology Guwahati, India

B.Tech. in Electronics and Electrical Engineering — CPI: 7.51 / 10.0

AUGUST 2017 - 2021

Experience

ConcertAI (Terarecon Team) — Bengaluru, India

SENIOR DATA SCIENTIST — APRIL 2025 - Present

- Built a low-resolution whole-heart segmentation and tubular-structure centerline extraction pipeline using 3D **nnUNet**, TotalSegmentator labels, and graph-based post-processing (connected components, distance transforms, minimal-cost paths, smoothing/resampling) on 3mm cardiac CT for efficient and robust multi-scan analysis.
- Built a class-conditioned latent diffusion pipeline for cardiac MRI slice synthesis using a shared **VAE** latent space and per-view diffusion model training; replaced text embeddings with anatomical segmentation-mask conditioning for structure-aware image generation.
- Developed a Cardiac MRI view classification model using pretrained **DINOv2**; performed embedding-based linear probing and progressive fine-tuning of transformer layers, using latent diffusion-based synthetic data generation for class imbalance mitigation.

Affogato.ai — Bengaluru, India

APPLIED AI ENGINEER — DECEMBER 2024 - MARCH 2025

- Developed a **Multi-LLM agent** chatbot that replaces traditional website UI with user prompts, enabling seamless interaction with the RenderNet public API. Utilized **MCP** (Model Context Protocol) by Anthropic and **LangGraph** to orchestrate multi-agent workflows.
- Developed an object placement workflow in **ComfyUI**, allowing users to seamlessly integrate objects into backgrounds. Utilized **Flux Inpainting** and **Flux Redux** to ensure high-fidelity detail preservation, natural blending, and scene consistency.

Qure.ai — Bengaluru, India

AI SCIENTIST — JUNE 2021 - NOVEMBER 2024

- Developed Segment-Anything Model **(MedSAM)** to segment pulmonary nodules on chest CT scans by integrating both user clicks based segmentation and bounding box prompts from a detection model.
- Developed a Large Language Model **(LLM)** similar to LLaMA2-7B, which generated detailed reports based on radiologist instructions, improving interpretative capabilities of vision model predictions.
- Implemented a self-supervised pre-training process using Masked Autoencoders **(MAE)** with Vision Transformers **(ViTs)** to extract latent representations, which were subsequently fine-tuned for advanced classification/segmentation tasks.
- Deployed developed models at scale, handling thousands of scans utilizing tools like **Docker** and Kubernetes. The pipeline is designed to be deployed both on-premise and on AWS ECS.
- Developed an object detection model with **RetinaNet**, achieving **92% sensitivity** in identifying pulmonary nodules in chest CT scans.
- Developed a robust algorithm for automated nodule tracking across timepoints in Chest CT scans, enhancing malignancy detection. Utilized image registration techniques and a deep learning-based lung segmentation model. Presented an e-poster at **UKIO 2023**.
- Led the creation of a deep learning solution for identifying Large Vessel Occlusion (LVO) in CTA scans with **90% sensitivity** and **92% specificity**, showcased at **WSC 2022** and also featured in **ICCV CVAMD 2023**, recognized as the primary inventor in the granted patent.
- Improved the efficiency and performance of an **open-source** lung segmentation model through response-based **knowledge distillation** techniques.

Samsung R&D Institute Delhi — Remote

SUMMER INTERN | MENTOR: **Mr. Rahul Jain** — MAY 2020 - JUNE 2020

- Developed a user-friendly web application using Python framework **Flask** to enable seamless access to deep learning based CNN models.
- Integrated **FaceNet** for facial recognition and **YOLO** for object detection, along with a custom-built landmark recognition model, enriching the application's capabilities. Developed a comprehensive functionality to add, remove, and search people from the website, enhancing user experience.

Hanyang University — Seoul, Republic of Korea

SUMMER RESEARCH INTERN | MENTOR: **Dr. Hyunchul Shin** — MAY 2019 - JULY 2019

- Worked on a proof of concept (PoC) that simulates time taken by the data(model weights) transfer between Nvidia Deep Learning Accelerator (NVDLA) (its sub-units) and GDDR6 SDRAM (off-chip memory).

Publications & Patents

Mind the Clot: Automated LVO Detection on CTA Using Deep Learning

Published at **INTERNATIONAL CONFERENCE ON COMPUTER VISION WORKSHOP** (ICCV) — 2023

System and Method for Automatically Detecting Large Vessel Occlusion on a Computational Tomography Angiogram

Granted by **UNITED STATES PATENT AND TRADEMARK OFFICE** — 2024

Skills & Competencies

Programming Languages Python • C • C++ • SQL • LATEX • CUDA
Tools & Frameworks Pytorch • OpenCV • Docker • FastAPI • AWS • MongoDB • PostgreSQL • LangChain/LangGraph
Domain & Interest 2D/3D Computer Vision • Large Language Models • Generative AI • Representation Learning

Achievements

2022 **Oral Presentation**, WSC 2022 Singapore, Deep Learning Based LVO Detection on CT Angiography of Brain